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EXPERIMENT – 23

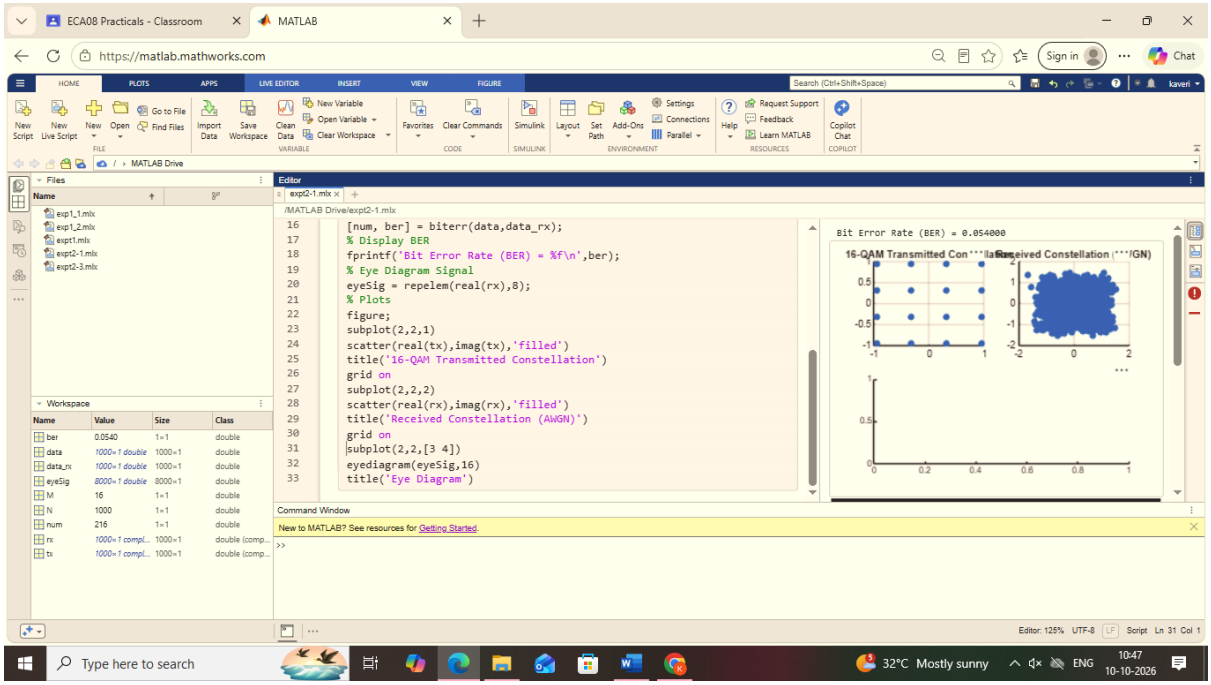
PERFORMANCE EVALUATION OF 8-QAM AND 16-QAM COMMUNICATION SYSTEMS UNDER AWGN USING BER, CONSTELLATION AND EYE DIAGRAM ANALYSIS IN MATLAB

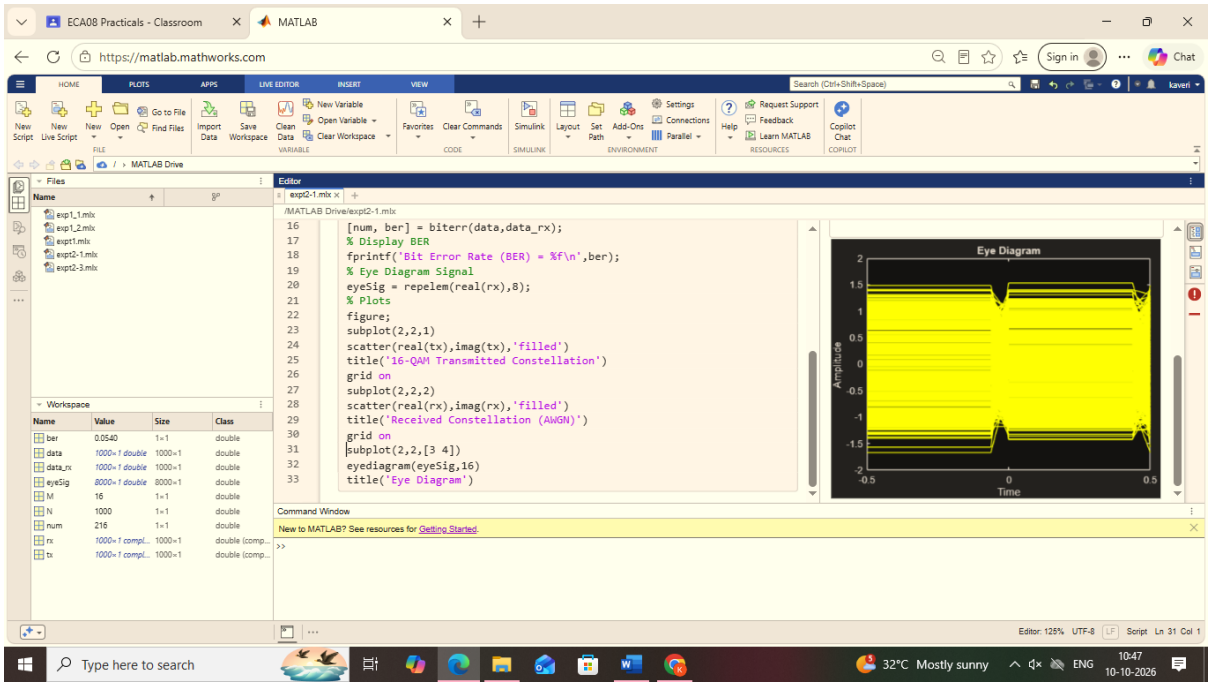
PROGRAM:

```
clc;
clear;
close all;
% Parameters
M = 16; % 16-QAM
N = 1000; % Number of symbols
data = randi([0 M-1],N,1);
% QAM Modulation
tx = qammod(data,M,'UnitAveragePower',true);
% AWGN Channel
rx = awgn(tx,10,'measured'); % 10 dB SNR
% Demodulation
data_rx = qamdemod(rx,M,'UnitAveragePower',true);
% BER Calculation
[num, ber] = biterr(data,data_rx);
% Display BER
fprintf('Bit Error Rate (BER) = %f\n',ber);
% Eye Diagram Signal
eyeSig = repelem(real(rx),8);
% Plots
figure;
subplot(2,2,1)
scatter(real(tx),imag(tx),'filled')
```

```
title('16-QAM Transmitted Constellation')
grid on
subplot(2,2,2)
scatter(real(rx),imag(rx),'filled')
title('Received Constellation (AWGN)')
grid on
subplot(2,2,[3 4])
eyediagram(eyeSig,16)
title('Eye Diagram')
```

OUTPUT:




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Files
exp1_1.mlx
exp1_2.mlx
expt1.mlx
expt2-1.mlx
expt2-3.mlx
Workspace
Name
Value
Size
Class
ber 0.0540 1×1 double
data 1000×1 double 1000×1 double
data_rx 1000×1 double 1000×1 double
eyeSig 8000×1 double 8000×1 double
M 16 1×1 double
N 1000 1×1 double
num 216 1×1 double
rx 1000×1 compl... 1000×1 double (comp...
tx 1000×1 compl... 1000×1 double (comp...
Editor
expt2-1.mlx
/MATLAB Drive/expt2-1.mlx
16 [num, ber] = biterr(data,data_rx);
17 % Display BER
18 fprintf('Bit Error Rate (BER) = %f\n',ber);
19 % Eye Diagram Signal
20 eyeSig = repelem(real(rx),8);
21 % Plots
22 figure;
23 subplot(2,2,1)
24 scatter(real(tx),imag(tx),'filled')
25 title('16-QAM Transmitted Constellation')
26 grid on
27 subplot(2,2,2)
28 scatter(real(rx),imag(rx),'filled')
29 title('Received Constellation (AWGN)')
30 grid on
31 subplot(2,2,[3 4])
32 eyediagram(eyeSig,16)
33 title('Eye Diagram')
Eye Diagram
Amplitude
Time
Command Window
New to MATLAB? See resources for Getting Started.
Type here to search
32°C Mostly sunny
ENG
10:47
10-10-2026